

NOTES AND CORRESPONDENCE

Tropical Cloudiness over the South Atlantic Ocean

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Westward moving cloud clusters have been noticed in the satellite cloud photographs over tropical latitudes. These cloud clusters are associated with westward moving wave disturbances. Chang (1970) applied a time-longitude technique for thin zonal strips to study these disturbances. Well defined bands of cloudiness which slope from east to west with increase of time could be seen in the time sections. The westward propagation speed could be calculated from the slope of the cloud lines. From the number of such lines during a period, the frequency could be calculated. Knowing the phase speed and frequency, the wavelength could be obtained. Thus, this technique could be very useful in obtaining the important characteristics of the disturbances over the tropical oceans, where the cloud picture is the only source of information.

In certain preferred locations, in certain periods of the year, a few of these westward moving tropical disturbances transform into violent tropical storms. The essential mechanism for development of the disturbances appears to have been successfully explained by the CISK (Conditional

Instability of the Second Kind) model (Charney and Eliassen, 1964 and Ooyama, 1964). However, it is not yet clear what aspect of the environment or model accounts for the fact that only a very few of the disturbances reach an advanced stage of development.

The situation in the South Atlantic Ocean is different in that in no season do the disturbances develop into tropical cyclones or even into intense tropical disturbances. Then the questions that arise are: 1) are there tropical disturbances in the South Atlantic similar to their counterparts in other tropics; 2) if they do exist, what aspect of the environment inhibits their development?

In the present note we report some preliminary results concerning the first question. We made use of the time longitude sections prepared by Wallace (1970) for the year 1967. These are the 5° latitude strips of clouds for 20°N to 20°S. We concentrate our attention on the cloudiness over South Atlantic Ocean.

2. Travelling disturbances

The existence of these disturbances has been suspected over the South Atlantic Ocean for quite some time. Fig. 1, adapted from Berry *et al.*

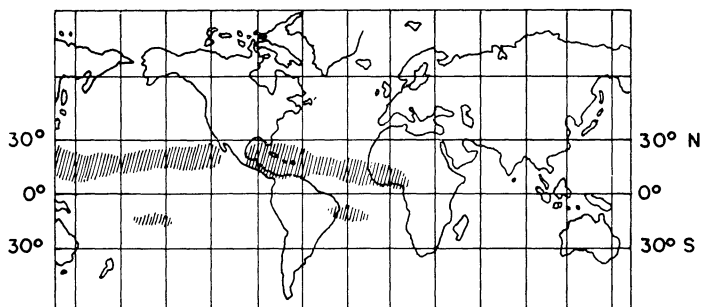


Fig. 1 Observed or suspected regions of westward travelling disturbances; adapted from Berry, Bollay and Beers (1945).

(1945), shows the observed or suspected regions of westward travelling disturbances; one of the regions being the tropical South Atlantic, extending into the northeast coast of Brazil. A careful examination, of Wallace's time section of clouds, revealed the existence of slanting lines of cloud

over the South Atlantic only during winter (June to August). In the $0-5^{\circ}\text{S}$ section, clouding appears to be less over South Atlantic. However, in $5-10^{\circ}\text{S}$ and $10-15^{\circ}\text{S}$ sections, distinct lines of cloudiness could be traced over the South Atlantic extending, into the Brazilian coast, up

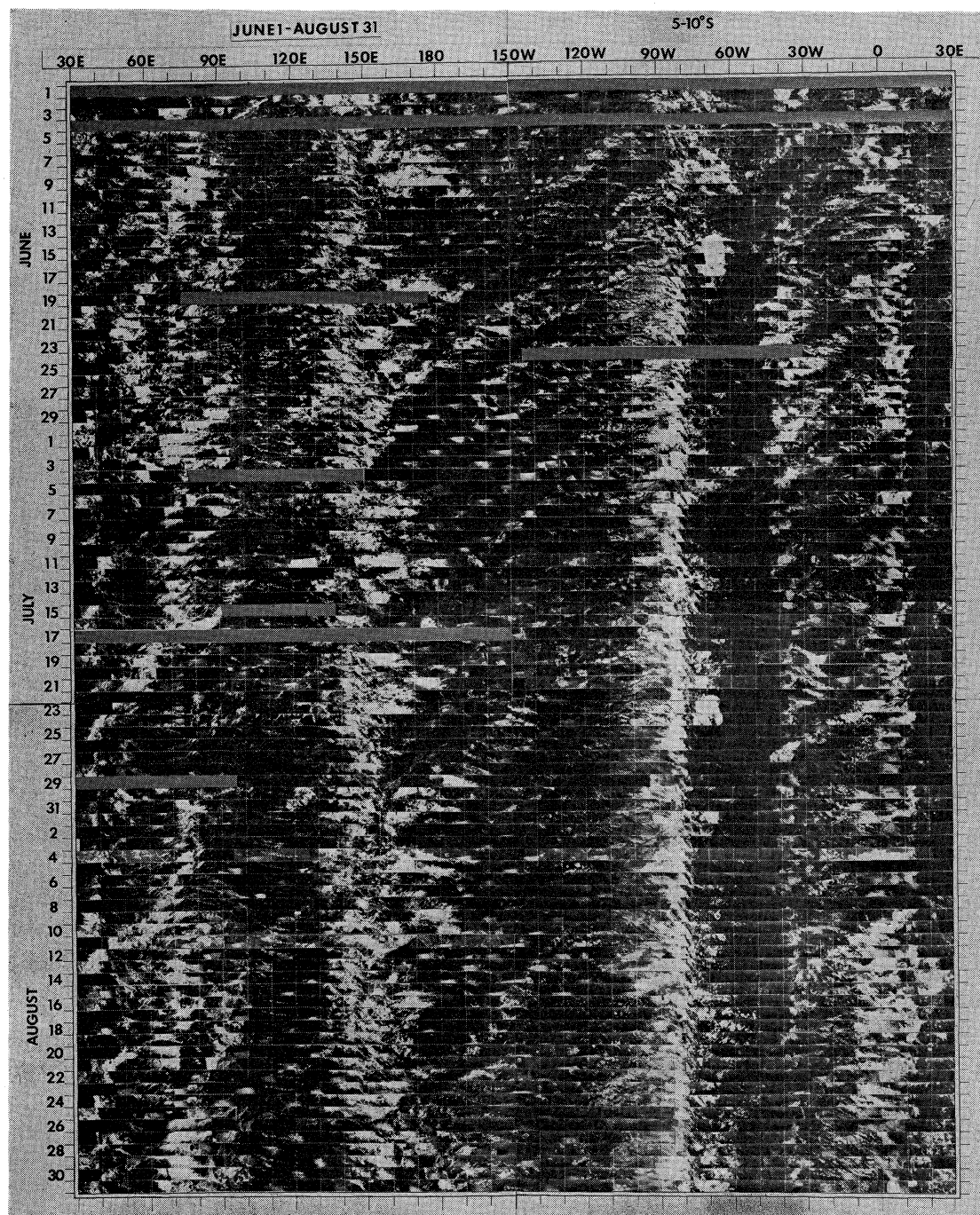


Fig. 2 Time-section of clouds during June 1-August 31, 1967 over $5-10^{\circ}\text{S}$; adapted from Wallace (1970).

to about 40°W . In all sections, a zone of permanent cloudiness could be seen around 80°W , off the west coast of South America. Sea surface temperatures are very low in this region and the cloudiness is probably associated with low stratus clouds or fog. Slanting lines of clouds could be seen over the South Pacific and in a few cases, they can be traced as far back as 10°E . A zone of permanent cloudiness could be seen around 10°E in the $0\text{--}5^{\circ}\text{S}$ section. This cloudiness might also be associated with low stratus or fog off the west coast of Africa. In other sections cloudiness around 10°E appears not to be persistent and, as mentioned above, slanting lines of clouds could be seen over the South Atlantic, starting around 10°E . Fig. 2 shows the $5\text{--}10^{\circ}\text{S}$ section adapted from Wallace (loc. cit.). Cloudiness appears to dissipate over the South American Continent, except for one or two lines of cloudiness inland and for some cloudiness very near the east coast.

It has been pointed out, by Wallace and Chang (1972), that the spectral analysis of the cloudiness is not particularly well suited for the study of tropical wave disturbances. A more successful approach, they pointed out, is direct visual inspection. Thus, we tried to infer some characteristics of the disturbances associated with cloud lines by visual inspection of time longitude sections. Using $5\text{--}10^{\circ}\text{S}$ strip (well defined lines of cloudiness could be seen in this section) it has been found that the periodicity of the associated disturbances over South Atlantic is several days (about 20 lines of clouding in 3 months). The mean speed of propagation appeared to be about 10 m-sec^{-1} (about 40° of longitude in 4 days). Thus the wavelength of the disturbances would be of the order of a few thousand kilometers.

3. Travelling disturbances and winter rainfall over Northeast Brazil

As mentioned earlier, the slanting lines of cloudiness could be traced from the African coast up to the Brazilian coast. In many cases, these clouds seem to extend a little inland. An examination of the cloud photographs for other winters also suggests this. It is interesting to note

that winter is the rainy season, for the coastal parts of northeast Brazil (Hsu and Wallace, 1976). This suggests that the winter rainfall along the coast appears to be associated with the westward moving disturbances from the Atlantic discussed above. Winter rainfall is confined to a very narrow zone along the coast and the region to the west of this zone is the well known Brazilian semi-arid area. A further study regarding the association of the coastal rainfall and the moving disturbances should await more data, particularly from the South Atlantic.

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References

- Berry, F. A., E. Bollay and Norman, R. Beers, 1945: Handbook of Meteorology, McGraw-Hill Book Company, p.1068.
- Chang, C. P., 1970: Westward Propagating Cloud Patterns in the Tropical Pacific as seen from Time-Composit Satellite Photographs. *J. Atmos. Sci.*, **27**, 133–138.
- Charney, J. G. and A. Eliassen, 1964: On the growth of the Hurricane Depression. *J. Atmos. Sci.*, **21**, 68–75.
- Hsu, Chin-Ping F. and J. W. Wallace, 1976. The Global distribution of Annual and Semiannual Cycles in Precipitation. *Mon. Wea. Rev.*, **104**, 1093–1101.
- Ooyama, K., 1964: A Dynamical Model for the Study of Tropical Cyclone Development. *Geofisica Intern.* (Mexico), **4**, 187–198.
- Wallace, J. M., 1970: Time-longitude Sections of Tropical Cloudiness December 1966–November 1967. ESSA Tech. Rept., NES-56, National Environmental Satellite Center, Washington, D.C., 37 pp.
- , and L. A. Chang, 1972: On the Application of Satellite Data and Cloud Brightness to the Study of Tropical Disturbances. *J. Atmos. Sci.*, **29**, 1400–1403.

南太平洋上の熱帯雲分布について

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